cancers

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Supplementary Materials: De Novo Carcinoma after Solid Organ Transplantation to Give Insight in Carcinogenesis in General, A Systematic Review and Meta-Analysis

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Table S1. Search strategy.

Pubmed September 10th 2020		
Search	**Query**	**Items found**
#8	Search (#7 NOT (animals[mh] NOT humans[mh]))	3516
#7	Search (#6 NOT (("Adolescent"[Mesh] OR "Child"[Mesh] OR "Infant"[Mesh] OR adolescen*[tiab] OR child*[tiab] OR schoolchild*[tiab] OR infant*[tiab] OR girl*[tiab] OR boy*[tiab] OR teen[tiab] OR teens[tiab] OR teenager*[tiab] OR youth*[tiab] OR pediatr*[tiab] OR paediatr*[tiab] OR puber*[tiab]) NOT ("Adult"[Mesh] OR adult*[tiab] OR man[tiab] OR men[tiab] OR woman[tiab] OR women[tiab])))	3546
#6	Search #4 and #5	3737
#5	Search ("Epidemiologic Studies"[Mesh] OR cohort[tiab] OR (case[tiab] AND (control[tiab] OR controll*[tiab] OR comparison[tiab] OR referent[tiab])) OR risk[tiab] OR causation[tiab] OR causal[tiab] OR "odds ratio"[tiab] OR etiol*[tiab] OR aetiol*[tiab] OR "natural history"[tiab] OR predict*[tiab] OR prognos*[tiab] OR outcome[tiab] OR course[tiab] OR retrospect*[tiab] OR followup[tiab] OR "follow up"[tiab])	7,007,005
#4	Search #1 AND #2 AND #3	5608
#3	Search "Carcinoma"[Mesh] OR "Sarcoma"[Mesh] OR "Melanoma"[Mesh] OR adenocarcinoma*[tiab] OR carcinoma*[tiab] OR sarcoma*[tiab] OR melanoma*[tiab] OR malign*[tiab]	1,628,750
#2	Search ("Organ Transplantation"[Mesh] OR transplant*[tiab] OR graft*[tiab] OR allotransplant*[tiab] OR allograft*[tiab] OR posttransplant*[tiab]) AND (heart[tiab] OR kidney[tiab] OR renal[tiab] OR liver[tiab] OR lung[tiab] OR solid organ*[tiab])	280,522
#1	Search "Immunosuppression"[Mesh] OR "Immunosuppressive Agents"[Mesh] OR immunosuppress*[tiab] OR immuno-suppress*[tiab] OR immunesuppress*[tiab] OR immune-suppress*[tiab] OR immunodepress*[tiab] OR immunedepress*[tiab] OR immuno-depress*[tiab] OR immune-depress*[tiab] OR allotype suppress*[tiab] OR antibody induction[tiab]	260,506

Embase September 10th 2020		
Search	**Query**	**Items found**
#9	#8 NOT (('adolescent'/exp OR 'child'/exp OR adolescent*:ti,ab OR child*:ti,ab OR schoolchild*:ti,ab OR infant*:ti,ab OR girl*:ti,ab OR boy*:ti,ab OR teen:ti,ab OR teens:ti,ab OR teenager*:ti,ab OR youth*:ti,ab OR pediatr*:ti,ab OR paediatr*:ti,ab OR puber*:ti,ab) NOT ('adult'/exp OR 'aged'/exp OR 'middle aged'/exp OR adult*:ti,ab OR man:ti,ab OR men:ti,ab OR woman:ti,ab OR women:ti,ab))	5561
#8	#6 AND #7	5915
#7	'epidemiology'/de OR cohort:ti,ab,kw OR (case:ti,ab,kw AND (control:ti,ab,kw OR controll*:ti,ab,kw OR comparison:ti,ab,kw OR referent:ti,ab,kw)) OR risk:ti,ab,kw OR causation:ti,ab,kw OR causal:ti,ab,kw OR 'odds ratio':ti,ab,kw OR etiol*:ti,ab,kw OR aetiol*:ti,ab,kw OR 'natural history':ti,ab,kw OR predict*:ti,ab,kw OR prognos*:ti,ab,kw OR outcome:ti,ab,kw OR course:ti,ab,kw OR retrospect*:ti,ab,kw OR followup:ti,ab,kw OR 'follow up':ti,ab,kw	8,819,132
#6	#5 NOT ([animals]/lim NOT [humans]/lim)	9477
#5	#4 NOT (conference*:it OR letter*:it OR editorial*:it)	9804
#4	#1 AND #2 AND #3	15,194
#3	'carcinoma'/exp OR 'sarcoma'/exp OR 'melanoma'/exp OR adenocarcinoma*:ab,ti,kw OR carcinoma*:ab,ti,kw OR sarcoma*:ab,ti,kw OR melanoma*:ab,ti,kw OR malign*:ab,ti,kw	2,365,914
#2	'solid organ transplantation'/exp OR 'solid organ transplant'/exp OR 'solid organ transplant recipient'/exp OR (('organ transplantation'/exp OR transplant*:ab,ti,kw OR graft*:ab,ti,kw OR allotransplant*:ab,ti,kw OR allograft*:ab,ti,kw OR posttransplant*:ab,ti,kw) AND (heart:ab,ti,kw OR kidney:ab,ti,kw OR renal:ab,ti,kw OR liver:ab,ti,kw OR lung:ab,ti,kw OR 'solid organ*':ab,ti,kw))	455,808

#1	'immunosuppressive treatment'/exp OR 'immunosuppressive agent'/exp OR immunosuppress*:ab,ti,kw OR 'immuno-suppress*':ab,ti,kw OR immunesuppress*:ab,ti,kw OR 'immune-suppress*':ab,ti,kw OR immunode-press*:ab,ti,kw OR immunedepress*:ab,ti,kw OR 'immuno-depress*':ab,ti,kw OR 'immune-depress*':ab,ti,kw OR 'allotype suppress*':ab,ti,kw OR 'antibody induction':ab,ti,kw	1,246,113

Table S2. Study characteristics * Follow up after transplantation (months): mean unless marked with † (median). ** Only reported for the patients with post-transplant malignancy. Abbreviations: NR: Not reported | USRDS: United States Renal Data System | OPTN: Organ Procurement and Transplantation Network | UNOS: United Network for Organ Sharing | ANZLTR: Australian & New Zealand Liver Transplant Registry | ANZCOTR: Australian and New Zealand Cardiotho-racic Organ Transplant Registry | CST: Canadian Society of Transplantation | ANZDATA: Australian and New Zealand Dialysis and Transplant registry | SRTR: Scientific Registry of Transplant Recipients.

Author	Country	Cohort	Total trans-plantations	Type of transplan-tation	Followup (months)*	Patients with de novo carcinoma, n (%)**	Type of carcinoma	Male (%)**
Aigner et al.**[1]**	Austria (1986–2005)	Hospital-based	3595	Solid organ	88	206 (5,73)	Any	76,9
Albright et al.[2]	US (1998–2001)	Hospital-based	402	Liver	62,4†	77 (19,15)	Any	NR
Arichi et al.[3]	Japan (1973–2007)	Hospital-based	429	Kidney	202	57 (13,29)	Any	63,2
Ativitavas et al.[4]	Thailand (1992–2007)	Hospital-based	270	Kidney	NR	18 (6,67)	Any	61,1
Baccarani et al.[5]	Italy (1991–2005)	Multicentre	417	Liver	81,6†	43 (10,31)	Any	74,4
Bruschi et al.[6]	Italy (1995–2011)	Hospital-based	878	Heart	135,6	22 (2,51)	Lung	90,9
Campistol et al.[7]	Multiple countries (1986–2005)	Multicentre	430	Kidney	60	17 (3,95)	Any	NR
Carenco et al. [8]	France (1991–2008)	Hospital-based	465	Liver	96	43 (9,25)	Any	NR
Chambade et al.[9]	France (2002–2006)	Hospital-based	2050	Kidney	17,4	7 (0,34)	Kidney	NR
Chang et al.[10]	Taiwan (2000–2015)	Hospital-based	222	Kidney	104	29 (13,1)	Bladder	NR
Chen et al.[11]	Taiwan (1992–2008)	Hospital-based	102	Kidney	45,5	9 (8,82)	Any	NR
Chen et al.[12]	Taiwan (2000–2012)	Multicentre	460	Kidney	75	31 (6,74)	Any	NR
Couetil et al.[13]	UK (1979–1988)	Hospital-based	303	Heart	NR	11 (3,63)	Any	100
Cox et al.[14]	US (1990–2010)	Hospital-based	5920	Kidney	45	11 (0,19)	Bladder	81,8
Curtil et al.[15]	France (1987–1996)	Hospital-based	267	Heart	NR	18 (6,74)	Any	94,4
Davis et al.[16]	Ireland (1976–2013)	Hospital-based	3741	Kidney	94	14 (0,37)	Bladder	NR
Demir et al.[17]	Turkey (2004–2009)	Hospital-based	100	Kidney	60	14 (14,00)	Any	NR
Doublet et al.[18]	France (1993–1995)	Hospital-based	129	Kidney	NR	5 (3,88)	Kidney	40
Eccher et al.[19]	Italy (1968–2015)	Hospital-based	1646	Kidney	100,8	16 (0,97)	Kidney	NR

Egeli et al.[20]	Turkey (1998–2016)	Hospital-based	429	Liver	103,4	9 (2,10)	Any	100
Elkentaoui et al.[21]	France (1983–2008)	Hospital-based	1350	Kidney	NR	36 (2,67)	Bladder	NR
Filocamo et al.[22]	Italy (1991–2007)	Hospital-based	694	Kidney	41	10 (1,44)	Kidney	70
Fox et al.[23]	Israel (1998–2012)	Hospital-based	412	Lung	60	39 (9,47)	Any	64,1
Frezza et al.[24]	US (1982–1992)	Hospital-based	1657	Liver	NR	50 (3,02)	Any	74
Gallagher et al.[25]	Australia (1983–1986)	Multicentre	481	Kidney	247,2†	226 (46,99)	Any	NR
Heinz-Peer et al.[26]	Austria (1993–1994)	NR	385	Kidney	36	6 (1,56)	Kidney	100
Hurst et al.[27]	US (2000–2005)	USRDS	40,281	Kidney	36	368 (0,91)	Kidney	66,3
Jonas et al.[28]	Germany (1988–1994)	Hospital-based	458	Liver	50,0†	33 (7,21)	Any	48,5
Kahan et al.[29]	US (1993–2002)	Hospital-based	1008	Kidney	45,4	35 (3,47)	Any	85,7
Kamal et al.[30]	Egypt (1976–2006)	Hospital-based	1865	Kidney	78	7 (0,38)	Bladder	100
Kanaan et al.[31]	Belgium (1976–2011)	Hospital-based	2944	Kidney	60	14 (0,48)	Liver	85,7
Karczewski et al.[32]	Poland (1994–2011)	Hospital-based	836	Kidney	120	63 (7,54)	Any	NR
Kauffman et al.[33]	US (1996–2001)	OPTN/UNOS	33,249	Kidney	32,1	569 (1,71)	Any	71,4
Kehinde et al.[34]	UK (1975–1991)	Hospital-based	492	Kidney	72	27 (5,49)	Any	66,7
Kellerman et al.[35]	US (1994–2007)	Hospital-based	851	Heart	64	73 (8,58)	Any	83,6
Keown et al.[36]	Multiple countries (1992–1994)	Multicentre	497	Kidney	12	44 (8,85)	Any	NR
Kim et al.[37]	Korea (1975–1999)	Multicentre	10,029	Kidney	156,2	193 (1,92)	Any	56
Kim et al.[38]	US (1995–2005)	Hospital-based	539	Kidney	120	18 (3,34)	Any	NR
Kwak et al.[39]	Korea (1999–2011)	Hospital-based	2139	Kidney	144	142 (6,64)	Any	NR
Lebkowska et al.[40]	Poland (NR)	Hospital-based	44	Kidney	NR	5 (11,36)	Thyroid	NR
Li et al.[41]	China (1998–2006)	Multicentre	1612	Kidney	57	9 (0,56)	Bladder	22,2
Lichtenberg et al.[42]	Israel (2001–2014)	Hospital-based	718	Kidney	42,8	32 (4,46)	Any	68,8
Liu et al.[43]	China (2002–2009)	Hospital-based	722	Kidney	NR	14 (1,94)	Any	78,6

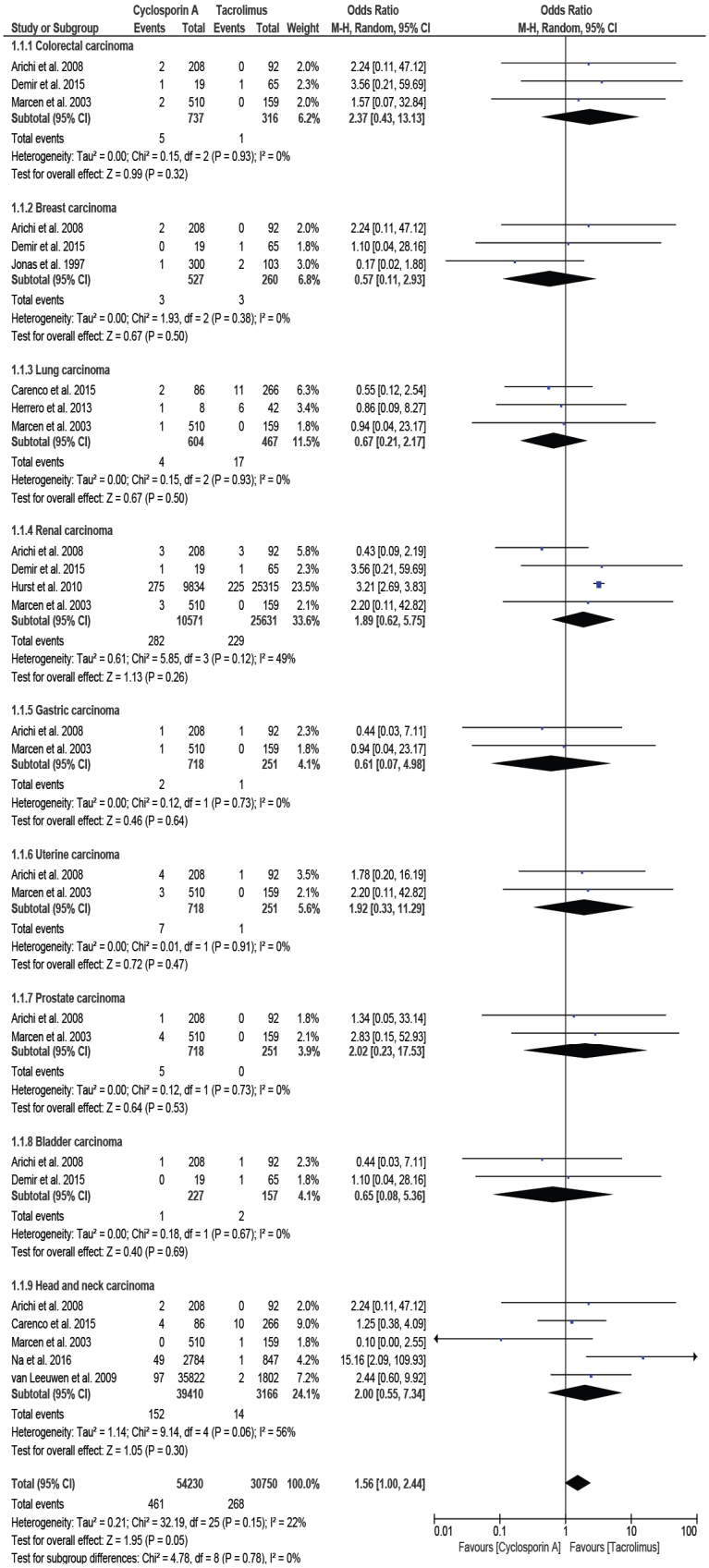
Lopez-Pintor et al.[44]	Spain (1989–2007)	Hospital-based	500	Kidney	59,7	6 (1,20)	Lip	100
Marcen et al.[45]	Spain (1979–2001)	Hospital-based	793	Kidney	75,4	95 (11,98)	Any	NR
Mathew et al.[46]	Multiple countries (NR)	Multicentre	1675	Kidney	24	44 (2,63)	Any	NR
McGeown et al.[47]	Ireland (1968–1998)	Hospital-based	868	Kidney	122,4	86 (9,91)	Any	NR
Mizuno et al.[48]	Japan (2002–2017)	Hospital-based	97	Liver	82,8	11 (11,34)	Any	90,9
Na et al.[49]	Australia (1984–2006)	ANZLTR, ANZCOTR	4141	Solid organ	63,0†	58 (1,40)	Lip	77,6
Neuzillet et al.[50]	France (1987–2003)	Hospital-based	933	Kidney	39	11 (1,18)	Kidney	90,9
Nure et al.[51]	Italy (2002–2012)	Hospital-based	225	Liver	NR	11 (4,89)	Any	NR
Oezcelik et al.[52]	Germany (2000–2009)	Hospital-based	761	Liver	70,3	5 (0,66)	Esophagus	80
Opelz et al.[53]	Multiple countries (1999–2013)	CST	4279	Kidney	50,4	374 (8,74)	Any	NR
Palazzetti et al.[54]	Italy (1988–2014)	Multicentre	3111	Kidney	NR	28 (0,90)	Bladder	67,9
Park et al.(1)[55]	Korea (1998–2008)	Hospital-based	1952	Liver	42	44 (2,25)	Any	79,5
Park et al.(2)[56]	Korea (2004–2007)	Hospital-based	509	Kidney	137	15 (2,95)	Stomach	66,7
Park et al.[57]	Korea (1991–2016)	Hospital-based	9776	Solid organ	240	13 (0,13)	Head and neck	92,3
Ploussard et al.[58]	France (1984–2006)	Hospital-based	2396	Kidney	43	12 (0,50)	Kidney	41,7
Raeisi et al.[59]	Iran (1991–2012)	Hospital-based	1487	Kidney	NR	64 (4,30)	Any	NR
Rinaldi et al.[60]	Italy (1985–1998)	Hospital-based	474	Heart	71,1	55 (11,60)	Any	89,1
Saigal et al.[61]	UK (1988–1999)	Hospital-based	1140	Liver	70,9	29 (2,54)	Any	72,4
Schmidt et al.[62]	Germany (1968–1994)	Hospital-based	868	Kidney	41,8	12 (1,38)	Any	50
Shoji et al.[63]	Japan (1997–2002)	Hospital-based	554	Liver	56	5 (0,90)	Lung	80
Singh et al.[64]	India (NR)	NR	1500	Kidney	NR	31 (2,07)	Head and neck	NR
Slavis et al.[65]	US (1963–1969)	Hospital-based	14	Kidney	22,5	7 (50,00)	Any	NR
Snanoudj et al.[66]	France (1983–2004)	Hospital-based	1267	Kidney	149,0†	150 (11,84)	Any	NR
Sobieszczanska et al.[67]	Poland (1987–2011)	Hospital-based	324	Heart	276	29 (8,95)	Any	82,8

Stauch et al.[68]	Germany (1986 - 1991)	Hospital-based	160	Heart	60	11 (6,88)	Any	100
Sun et al.[69]	Korea (1991–2010)	Hospital-based	1425	Kidney	194,4	5 (0,35)	Kidney	60
Tillou et al.[70]	France (1988–2009)	Multicentre	41,806	Kidney	38,1	79 (0,19)	Kidney	68,4
Tiwari et al.[71]	India (2006–2017)	Hospital-based	2100	Liver	NR	21 (1,00)	Any	95
Tomaszewski et al.[72]	UK (1992–2007)	Hospital-based	5686	Solid organ	40	11 (0,19)	Bladder	100
van Leeuwen et al.[73]	US (1982–2003)	ANZDATA	8162	Kidney	NR	203 (2,49)	Lip	84,4
Vegso et al.[74]	Hungary (1973–2012)	Hospital-based	3530	Kidney	159,3	232 (6,57)	Any	55,6
Vogt et al.[75]	Germany (1981–1986)	Hospital-based	598	Kidney	41,8	18 (3,01)	Any	NR
Wang et al.[76]	Taiwan (1990–2015)	Hospital-based	454	Heart	69,2	27 (5,95)	Any	96,3
Wu et al.[77]	Taiwan (1983–2003)	Hospital-based	730	Kidney	72,2	63 (8,63)	Any	NR
Yanik et al.[78]	US (1995–2009)	SRTR, OPTN	5687	Kidney	25,4	85 (1,49)	Any	NR
Yilmaz et al.[79]	Turkey (2000–2015)	Hospital-based	867	Kidney	112,9	59 (6,81)	Any	69,5
Yoshimura et al.[80]	Japan (1970–1979)	Hospital-based	110	Kidney	300	7 (6,36)	Any	NR
Yserbyt et al.[81]	Belgium (2010–2012)	Hospital-based	494	Lung	126	13 (2,63)	Lung	69,2
Yu et al.[82]	China (2005–2011)	Hospital-based	569	Liver	44,9	16 (2,81)	Any	81,3

Table S3. Newcastle-Ottowa score.

Author	Year	Selection	Comparability	Outcome	NOS score
Aigner et al.[1]	2007	☆☆☆☆	☆	☆☆	7
Albright et al.[2]	2010	☆☆☆☆	☆	☆☆☆	8
Arichi et al.[3]	2008	☆☆☆	☆	☆☆	6
Ativitavas et al.[4]	2008	☆☆☆☆	☆	☆☆	7
Baccarani et al.[5]	2010	☆☆☆	☆	☆☆	6
Bruschi et al.[6]	2013	☆☆	☆	☆☆	5
Campistol et al.[7]	2006	☆☆☆☆	☆	☆☆	7
Carenco et al. [8]	2015	☆☆☆	☆	☆☆☆	7
Chambade et al.[9]	2008	☆☆	☆	☆☆☆	6
Chang et al.[10]	2019	☆☆☆☆	☆	☆☆☆	8
Chen et al.[11]	2009	☆☆☆	☆	☆☆	6
Chen et al.[12]	2015	☆☆☆☆	☆	☆☆☆	8
Couetil et al.[13]	1990	☆☆☆☆	☆	☆☆	7
Cox et al.[14]	2011	☆☆☆	☆	☆☆	6
Curtil et al.[15]	1997	☆☆☆	☆	☆☆☆	7
Davis et al.[16]	2013	☆☆☆	☆	☆☆☆	7
Demir et al.[17]	2015	☆☆☆☆	☆	☆☆☆	8
Doublet et al.[18]	1997	☆☆☆☆	☆	☆☆	7
Eccher et al.[19]	2016	☆☆☆☆	☆	☆☆☆	8
Egeli et al.[20]	2017	☆☆☆☆	☆	☆☆	7
Elkentaoui et al.[21]	2010	☆☆☆	☆	☆☆	6
Filocamo et al.[22]	2009	☆☆☆	☆	☆☆	6
Fox et al.[23]	2017	☆☆☆	☆	☆☆	6
Frezza et al.[24]	1997	☆☆☆☆	☆	☆	6
Gallagher et al.[25]	2010	☆☆☆	☆	☆☆☆	7
Heinz-Peer et al.[26]	1995	☆☆☆☆	☆	☆☆	7
Hurst et al.[27]	2010	☆☆☆	☆	☆☆	6
Jonas et al.[28]	1997	☆☆☆☆	☆	☆☆	7
Kahan et al.[29]	2005	☆☆☆	☆	☆☆	6
Kamal et al.[30]	2008	☆☆☆	☆	☆☆☆	7
Kanaan et al.[31]	2017	☆☆	☆	☆☆☆	6
Karczewski et al.[32]	2012	☆☆☆☆	☆	☆☆☆	8
Kauffman et al.[33]	2005	☆☆☆	☆	☆☆	6
Kehinde et al.[34]	1994	☆☆☆☆	☆	☆☆	7
Kellerman et al.[35]	2009	☆☆☆☆	☆	☆☆☆	8
Keown et al.[36]	1996	☆☆☆	☆	☆☆	6
Kim et al.[37]	2010	☆☆☆	☆	☆☆☆	7
Kim et al.[38]	2002	☆☆☆☆	☆	☆☆☆	8
Kwak et al.[39]	2013	☆☆☆☆	☆	☆☆☆	8
Lebkowska et al.[40]	2003	☆☆☆	☆	☆☆☆	7
Li et al.[41]	2009	☆☆☆	☆	☆☆☆	7
Lichtenberg et al.[42]	2017	☆☆☆	☆	☆	5
Liu et al.[43]	2015	☆☆☆☆	☆	☆☆	7
Lopez-Pintor et al.[44]	2011	☆☆☆☆	☆	☆☆	7
Marcen et al.[45]	2003	☆☆☆	☆	☆☆	6
Mathew et al.[46]	2004	☆☆☆	☆	☆	5
McGeown et al.[47]	2000	☆☆☆	☆	☆☆☆	7

Mizuno et al.[48]	2018	☆☆☆	☆	☆☆	6
Na et al.[49]	2016	☆☆☆	☆	☆☆	6
Neuzillet et al.[50]	2005	☆☆☆☆	☆	☆☆	7
Nure et al.[51]	2013	☆☆☆☆	☆	☆☆	7
Oezcelik et al.[52]	2011	☆☆☆☆	☆	☆☆	7
Opelz et al.[53]	2016	☆☆☆☆	☆	☆☆	7
Palazzetti et al.[54]	2018	☆☆☆☆	☆	☆	6
Park et al.(1)[55]	2012	☆☆☆	☆	☆	5
Park et al.(2)[56]	2012	☆☆☆☆	☆	☆☆	7
Park et al.[57]	2018	☆☆☆☆	☆	☆☆	7
Ploussard et al.[58]	2012	☆☆☆	☆	☆☆	6
Raeisi et al.[59]	2013	☆☆☆☆	☆	☆☆	7
Rinaldi et al.[60]	2001	☆☆☆☆	☆	☆☆☆	8
Saigal et al.[61]	2002	☆☆☆☆	☆	☆☆	7
Schmidt et al.[62]	1995	☆☆☆☆	☆	☆	6
Shoji et al.[63]	2017	☆☆	☆	☆☆	5
Singh et al.[64]	2006	☆☆☆	☆	☆☆	6
Slavis et al.[65]	1990	☆☆☆☆	☆	☆☆	7
Snanoudj et al.[66]	2004	☆☆☆☆	☆	☆☆	7
Sobieszczanska et al.[67]	2013	☆☆☆☆	☆	☆☆☆	8
Stauch et al.[68]	1993	☆☆☆☆	☆	☆☆	7
Sun et al.[69]	2013	☆☆	☆	☆☆☆	6
Tillou et al.[70]	2012	☆☆☆	☆	☆☆	6
Tiwari et al.[71]	2020	☆☆☆☆	☆	☆☆	7
Tomaszewski et al.[72]	2011	☆☆	☆	☆☆	5
van Leeuwen et al.[73]	2009	☆☆☆☆	☆	☆☆	7
Vegso et al.[74]	2011	☆☆☆☆	☆	☆☆	7
Vogt et al.[75]	1990	☆☆☆☆	☆	☆	6
Wang et al.[76]	2016	☆☆☆	☆	☆☆	6
Wu et al.[77]	2004	☆☆☆☆	☆	☆☆☆	8
Yanik et al.[78]	2015	☆☆☆	☆	☆	5
Yilmaz et al.[79]	2016	☆☆☆☆	☆	☆☆☆	8
Yoshimura et al.[80]	2005	☆☆☆	☆	☆☆☆	7
Yserbyt et al.[81]	2012	☆☆☆☆	☆	☆☆	7
Yu et al.[82]	2014	☆☆☆	☆	☆	5


Cyclosporin A
Tacrolimus
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
1.1.1 Colorectal carcinoma
Arichi et al. 2008 2 208 0 92 2.0% 2.24 [0.11, 47.12]
Demir et al. 2015 1 19 1 65 2.3% 3.56 [0.21, 59.69]
Marcen et al. 2003 2 510 0 159 2.0% 1.57 [0.07, 32.84]
Subtotal (95% CI) 737 316 6.2% 2.37 [0.43, 13.13]
Total events 5 1
Heterogeneity: Tau² = 0.00; Chi² = 0.15, df = 2 (P = 0.93); I² = 0%
Test for overall effect: Z = 0.99 (P = 0.32)
1.1.2 Breast carcinoma
Arichi et al. 2008 2 208 0 92 2.0% 2.24 [0.11, 47.12]
Demir et al. 2015 0 19 1 65 1.8% 1.10 [0.04, 28.16]
Jonas et al. 1997 1 300 2 103 3.0% 0.17 [0.02, 1.88]
Subtotal (95% CI) 527 260 6.8% 0.57 [0.11, 2.93]
Total events 3 3
Heterogeneity: Tau² = 0.00; Chi² = 1.93, df = 2 (P = 0.38); I² = 0%
Test for overall effect: Z = 0.67 (P = 0.50)
1.1.3 Lung carcinoma
Carenco et al. 2015 2 86 11 266 6.3% 0.55 [0.12, 2.54]
Herrero et al. 2013 1 8 6 42 3.4% 0.86 [0.09, 8.27]
Marcen et al. 2003 1 510 0 159 1.8% 0.94 [0.04, 23.17]
Subtotal (95% CI) 604 467 11.5% 0.67 [0.21, 2.17]
Total events 4 17
Heterogeneity: Tau² = 0.00; Chi² = 0.15, df = 2 (P = 0.93); I² = 0%
Test for overall effect: Z = 0.67 (P = 0.50)
1.1.4 Renal carcinoma
Arichi et al. 2008 3 208 3 92 5.8% 0.43 [0.09, 2.19]
Demir et al. 2015 1 19 1 65 2.3% 3.56 [0.21, 59.69]
Hurst et al. 2010 275 9834 225 25315 23.5% 3.21 [2.69, 3.83]
Marcen et al. 2003 3 510 0 159 2.1% 2.20 [0.11, 42.82]
Subtotal (95% CI) 10571 25631 33.6% 1.89 [0.62, 5.75]
Total events 282 229
Heterogeneity: Tau² = 0.61; Chi² = 5.85, df = 3 (P = 0.12); I² = 49%
Test for overall effect: Z = 1.13 (P = 0.26)
1.1.5 Gastric carcinoma
Arichi et al. 2008 1 208 1 92 2.3% 0.44 [0.03, 7.11]
Marcen et al. 2003 1 510 0 159 1.8% 0.94 [0.04, 23.17]
Subtotal (95% CI) 718 251 4.1% 0.61 [0.07, 4.98]
Total events 2 1
Heterogeneity: Tau² = 0.00; Chi² = 0.12, df = 1 (P = 0.73); I² = 0%
Test for overall effect: Z = 0.46 (P = 0.64)
1.1.6 Uterine carcinoma
Arichi et al. 2008 4 208 1 92 3.5% 1.78 [0.20, 16.19]
Marcen et al. 2003 3 510 0 159 2.1% 2.20 [0.11, 42.82]
Subtotal (95% CI) 718 251 5.6% 1.92 [0.33, 11.29]
Total events 7 1
Heterogeneity: Tau² = 0.00; Chi² = 0.01, df = 1 (P = 0.91); I² = 0%
Test for overall effect: Z = 0.72 (P = 0.47)
1.1.7 Prostate carcinoma
Arichi et al. 2008 1 208 0 92 1.8% 1.34 [0.05, 33.14]
Marcen et al. 2003 4 510 0 159 2.1% 2.83 [0.15, 52.93]
Subtotal (95% CI) 718 251 3.9% 2.02 [0.23, 17.53]
Total events 5 0
Heterogeneity: Tau² = 0.00; Chi² = 0.12, df = 1 (P = 0.73); I² = 0%
Test for overall effect: Z = 0.64 (P = 0.53)
1.1.8 Bladder carcinoma
Arichi et al. 2008 1 208 1 92 2.3% 0.44 [0.03, 7.11]
Demir et al. 2015 0 19 1 65 1.8% 1.10 [0.04, 28.16]
Subtotal (95% CI) 227 157 4.1% 0.65 [0.08, 5.36]
Total events 1 2
Heterogeneity: Tau² = 0.00; Chi² = 0.18, df = 1 (P = 0.67); I² = 0%
Test for overall effect: Z = 0.40 (P = 0.69)
1.1.9 Head and neck carcinoma
Arichi et al. 2008 2 208 0 92 2.0% 2.24 [0.11, 47.12]
Carenco et al. 2015 4 86 10 266 9.0% 1.25 [0.38, 4.09]
Marcen et al. 2003 0 510 1 159 1.8% 0.10 [0.00, 2.55]
Na et al. 2016 49 2784 1 847 4.2% 15.16 [2.09, 109.93]
van Leeuwen et al. 2009 97 35822 2 1802 7.2% 2.44 [0.60, 9.92]
Subtotal (95% CI) 39410 3166 24.1% 2.00 [0.55, 7.34]
Total events 152 14
Heterogeneity: Tau² = 1.14; Chi² = 9.14, df = 4 (P = 0.06); I² = 56%
Test for overall effect: Z = 1.05 (P = 0.30)
Total (95% CI) 54230 30750 100.0% 1.56 [1.00, 2.44]
Total events 461 268
Heterogeneity: Tau² = 0.21; Chi² = 32.19, df = 25 (P = 0.15); I² = 22%
Test for overall effect: Z = 1.95 (P = 0.05)
Test for subgroup differences: Chi² = 4.78, df = 8 (P = 0.78), I² = 0%
0.01 0.1 1 10 100
Favours [Cyclosporin A] Favours [Tacrolimus]

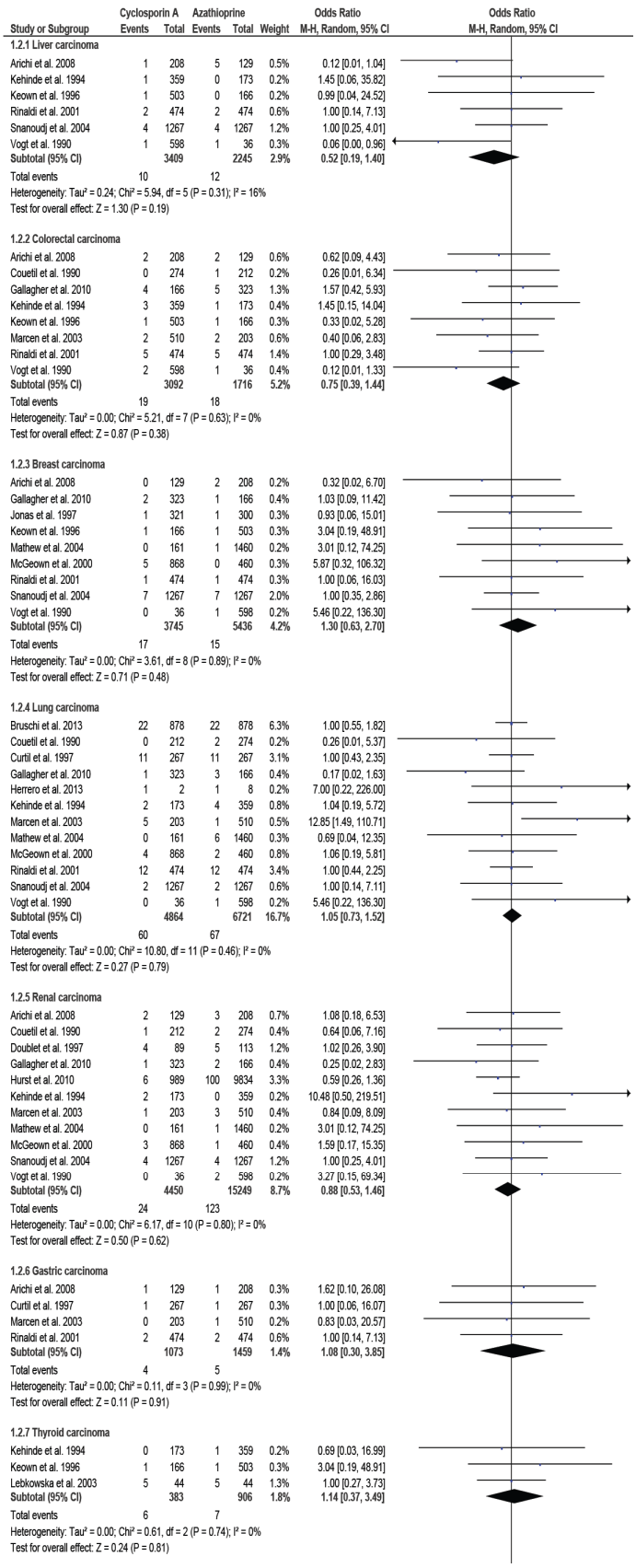


Figure S1. Cyclosporine A versus Tacrolimus.

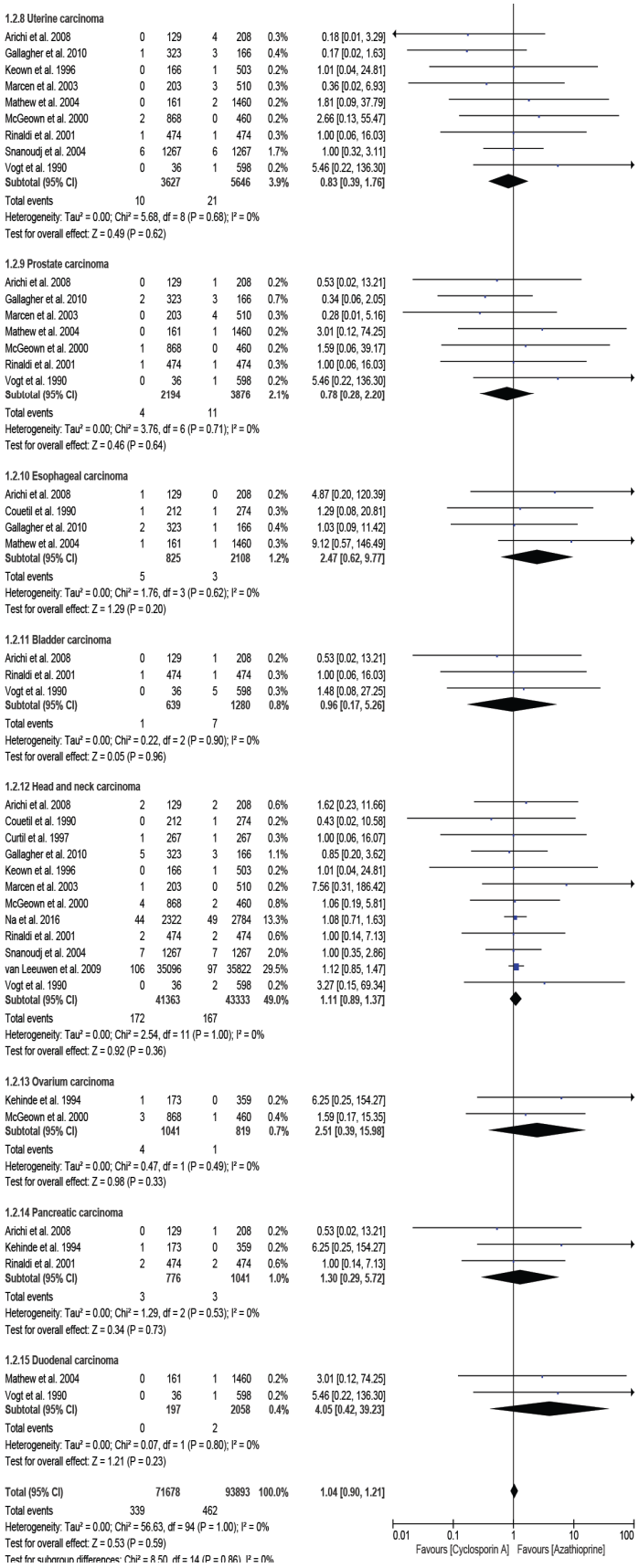

1.2.8 Uterine carcinoma
Arichi et al. 2008 0 129 4 208 0.3% 0.18 [0.01, 3.29]
Gallagher et al. 2010 1 323 3 166 0.4% 0.17 [0.02, 1.63]
Keown et al. 1996 0 166 1 503 0.2% 1.01 [0.04, 24.81]
Marcen et al. 2003 0 203 3 510 0.3% 0.36 [0.02, 6.93]
Mathew et al. 2004 0 161 2 1460 0.2% 1.81 [0.09, 37.79]
McGeown et al. 2000 2 868 0 460 0.2% 2.66 [0.13, 55.47]
Rinaldi et al. 2001 1 474 1 474 0.3% 1.00 [0.06, 16.03]
Snanoudj et al. 2004 6 1267 6 1267 1.7% 1.00 [0.32, 3.11]
Vogt et al. 1990 0 36 1 598 0.2% 5.46 [0.22, 136.30]
Subtotal (95% CI) 3627 5646 3.9% 0.83 [0.39, 1.76]
Total events 10 21
Heterogeneity: Tau² = 0.00; Chi² = 5.68, df = 8 (P = 0.68); I² = 0%
Test for overall effect: Z = 0.49 (P = 0.62)
1.2.9 Prostate carcinoma
Arichi et al. 2008 0 129 1 208 0.2% 0.53 [0.02, 13.21]
Gallagher et al. 2010 2 323 3 166 0.7% 0.34 [0.06, 2.05]
Marcen et al. 2003 0 203 4 510 0.3% 0.28 [0.01, 5.16]
Mathew et al. 2004 0 161 1 1460 0.2% 3.01 [0.12, 74.25]
McGeown et al. 2000 1 868 0 460 0.2% 1.59 [0.06, 39.17]
Rinaldi et al. 2001 1 474 1 474 0.3% 1.00 [0.06, 16.03]
Vogt et al. 1990 0 36 1 598 0.2% 5.46 [0.22, 136.30]
Subtotal (95% CI) 2194 3876 2.1% 0.78 [0.28, 2.20]
Total events 4 11
Heterogeneity: Tau² = 0.00; Chi² = 3.76, df = 6 (P = 0.71); I² = 0%
Test for overall effect: Z = 0.46 (P = 0.64)
1.2.10 Esophageal carcinoma
Arichi et al. 2008 1 129 0 208 0.2% 4.87 [0.20, 120.39]
Couetil et al. 1990 1 212 1 274 0.3% 1.29 [0.08, 20.81]
Gallagher et al. 2010 2 323 1 166 0.4% 1.03 [0.09, 11.42]
Mathew et al. 2004 1 161 1 1460 0.3% 9.12 [0.57, 146.49]
Subtotal (95% CI) 825 2108 1.2% 2.47 [0.62, 9.77]
Total events 5 3
Heterogeneity: Tau² = 0.00; Chi² = 1.76, df = 3 (P = 0.62); I² = 0%
Test for overall effect: Z = 1.29 (P = 0.20)
1.2.11 Bladder carcinoma
Arichi et al. 2008 0 129 1 208 0.2% 0.53 [0.02, 13.21]
Rinaldi et al. 2001 1 474 1 474 0.3% 1.00 [0.06, 16.03]
Vogt et al. 1990 0 36 5 598 0.3% 1.48 [0.08, 27.25]
Subtotal (95% CI) 639 1280 0.8% 0.96 [0.17, 5.26]
Total events 1 7
Heterogeneity: Tau² = 0.00; Chi² = 0.22, df = 2 (P = 0.90); I² = 0%
Test for overall effect: Z = 0.05 (P = 0.96)
1.2.12 Head and neck carcinoma
Arichi et al. 2008 2 129 2 208 0.6% 1.62 [0.23, 11.66]
Couetil et al. 1990 0 212 1 274 0.2% 0.43 [0.02, 10.58]
Curtil et al. 1997 1 267 1 267 0.3% 1.00 [0.06, 16.07]
Gallagher et al. 2010 5 323 3 166 1.1% 0.85 [0.20, 3.62]
Keown et al. 1996 0 166 1 503 0.2% 1.01 [0.04, 24.81]
Marcen et al. 2003 1 203 0 510 0.2% 7.56 [0.31, 186.42]
McGeown et al. 2000 4 868 2 460 0.8% 1.06 [0.19, 5.81]
Na et al. 2016 44 2322 49 2784 13.3% 1.08 [0.71, 1.63]
Rinaldi et al. 2001 2 474 2 474 0.6% 1.00 [0.14, 7.13]
Snanoudj et al. 2004 7 1267 7 1267 2.0% 1.00 [0.35, 2.86]
van Leeuwen et al. 2009 106 35096 97 35822 29.5% 1.12 [0.85, 1.47]
Vogt et al. 1990 0 36 2 598 0.2% 3.27 [0.15, 69.34]
Subtotal (95% CI) 41363 43333 49.0% 1.11 [0.89, 1.37]
Total events 172 167
Heterogeneity: Tau² = 0.00; Chi² = 2.54, df = 11 (P = 1.00); I² = 0%
Test for overall effect: Z = 0.92 (P = 0.36)
1.2.13 Ovarium carcinoma
Kehinde et al. 1994 1 173 0 359 0.2% 6.25 [0.25, 154.27]
McGeown et al. 2000 3 868 1 460 0.4% 1.59 [0.17, 15.35]
Subtotal (95% CI) 1041 819 0.7% 2.51 [0.39, 15.98]
Total events 4 1
Heterogeneity: Tau² = 0.00; Chi² = 0.47, df = 1 (P = 0.49); I² = 0%
Test for overall effect: Z = 0.98 (P = 0.33)
1.2.14 Pancreatic carcinoma
Arichi et al. 2008 0 129 1 208 0.2% 0.53 [0.02, 13.21]
Kehinde et al. 1994 1 173 0 359 0.2% 6.25 [0.25, 154.27]
Rinaldi et al. 2001 2 474 2 474 0.6% 1.00 [0.14, 7.13]
Subtotal (95% CI) 776 1041 1.0% 1.30 [0.29, 5.72]
Total events 3 3
Heterogeneity: Tau² = 0.00; Chi² = 1.29, df = 2 (P = 0.53); I² = 0%
Test for overall effect: Z = 0.34 (P = 0.73)
1.2.15 Duodenal carcinoma
Mathew et al. 2004 0 161 1 1460 0.2% 3.01 [0.12, 74.25]
Vogt et al. 1990 0 36 1 598 0.2% 5.46 [0.22, 136.30]
Subtotal (95% CI) 197 2058 0.4% 4.05 [0.42, 39.23]
Total events 0 2
Heterogeneity: Tau² = 0.00; Chi² = 0.07, df = 1 (P = 0.80); I² = 0%
Test for overall effect: Z = 1.21 (P = 0.23)
Total (95% CI) 71678 93893 100.0% 1.04 [0.90, 1.21]
Total events 339 462
Heterogeneity: Tau² = 0.00; Chi² = 56.63, df = 94 (P = 1.00); I² = 0%
Test for overall effect: Z = 0.53 (P = 0.59)
Test for subgroup differences: Chi² = 8.50, df = 14 (P = 0.86), I² = 0%
0.01 0.1 1 10 100
Favours [Cyclosporin A] Favours [Azathioprine]

Figure S2. Cyclosporine A versus Azathioprine.

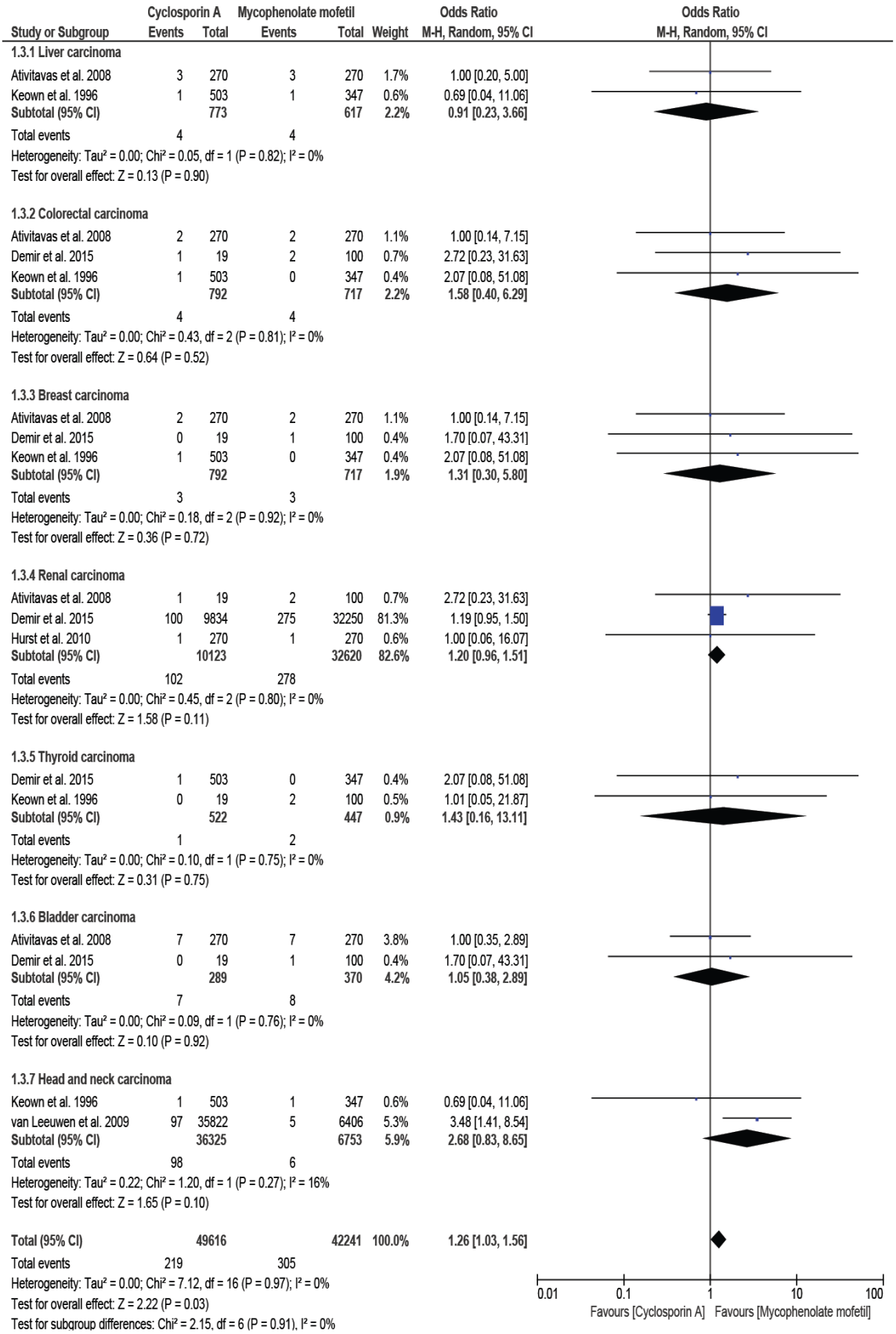


Figure S3. Cyclosporine A versus Mocyphenolate mofetil.

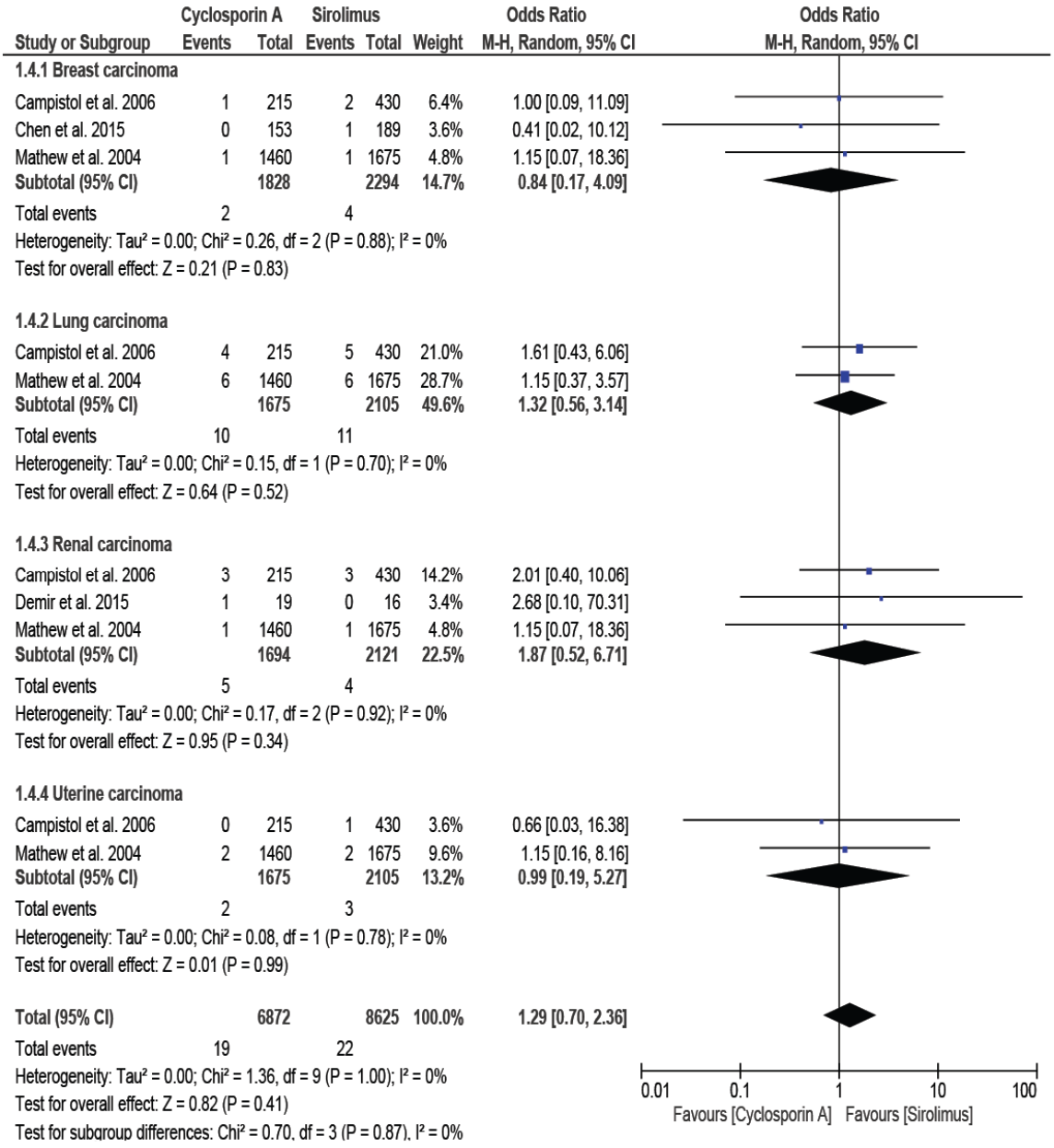


Figure S4. Cyclosporine A versus Sirolimus.

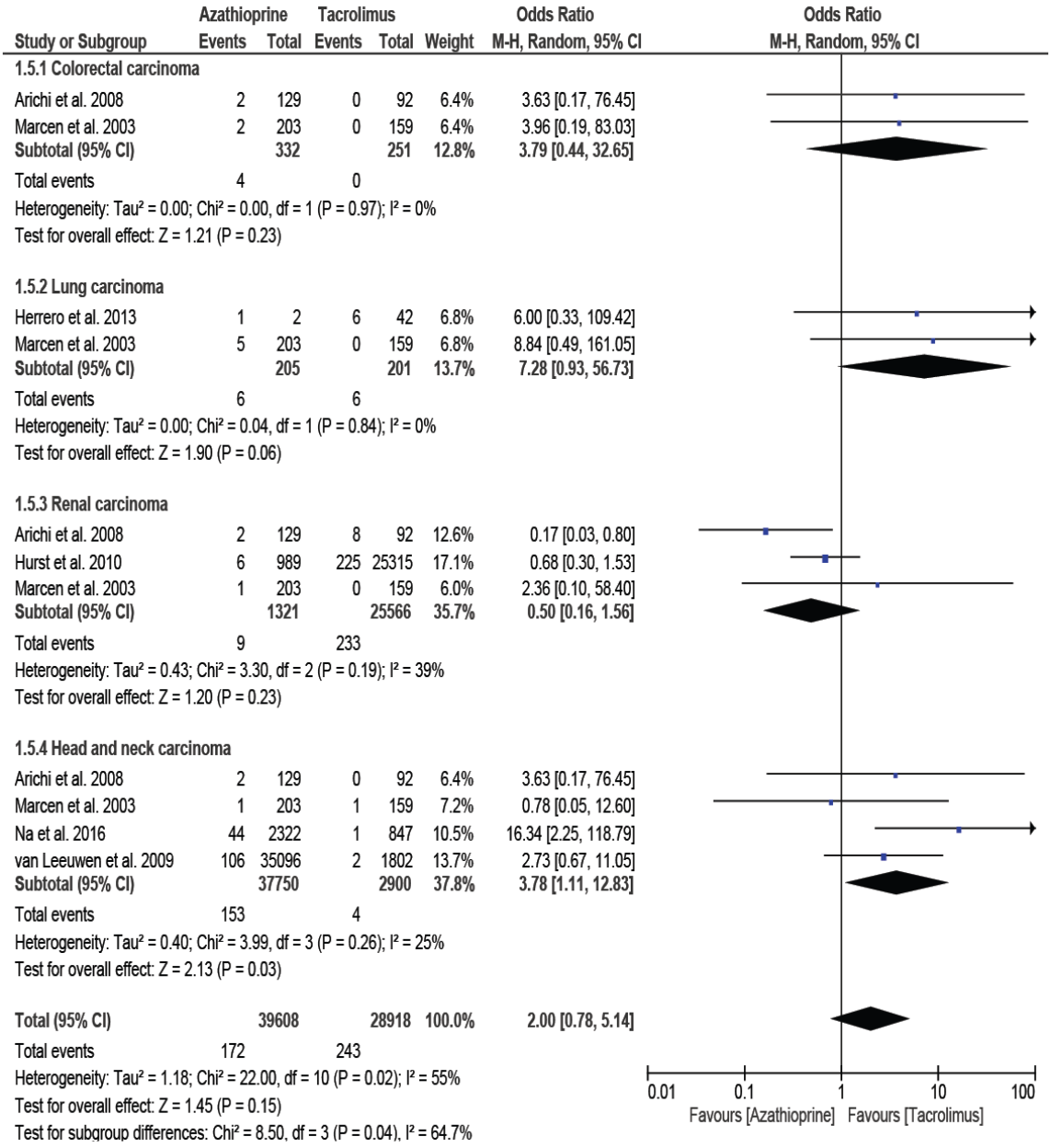


Figure S5. Azathioprine versus Tacrolimus.

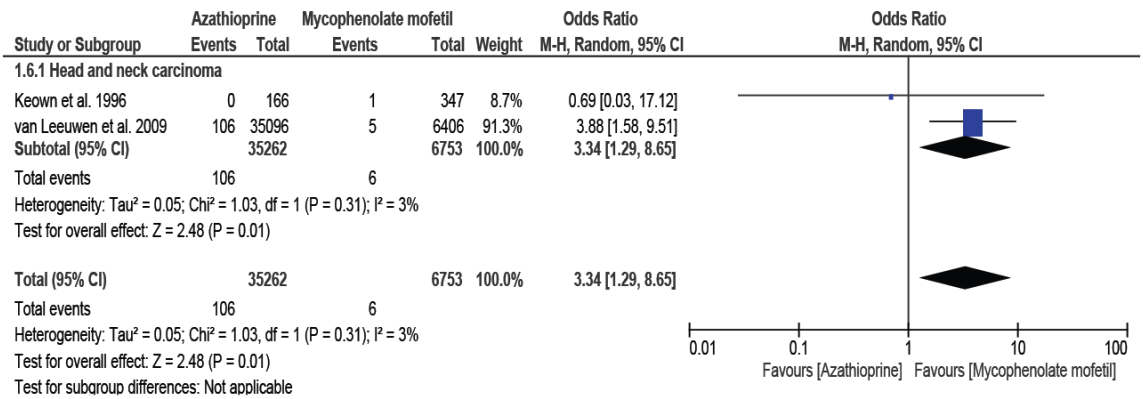
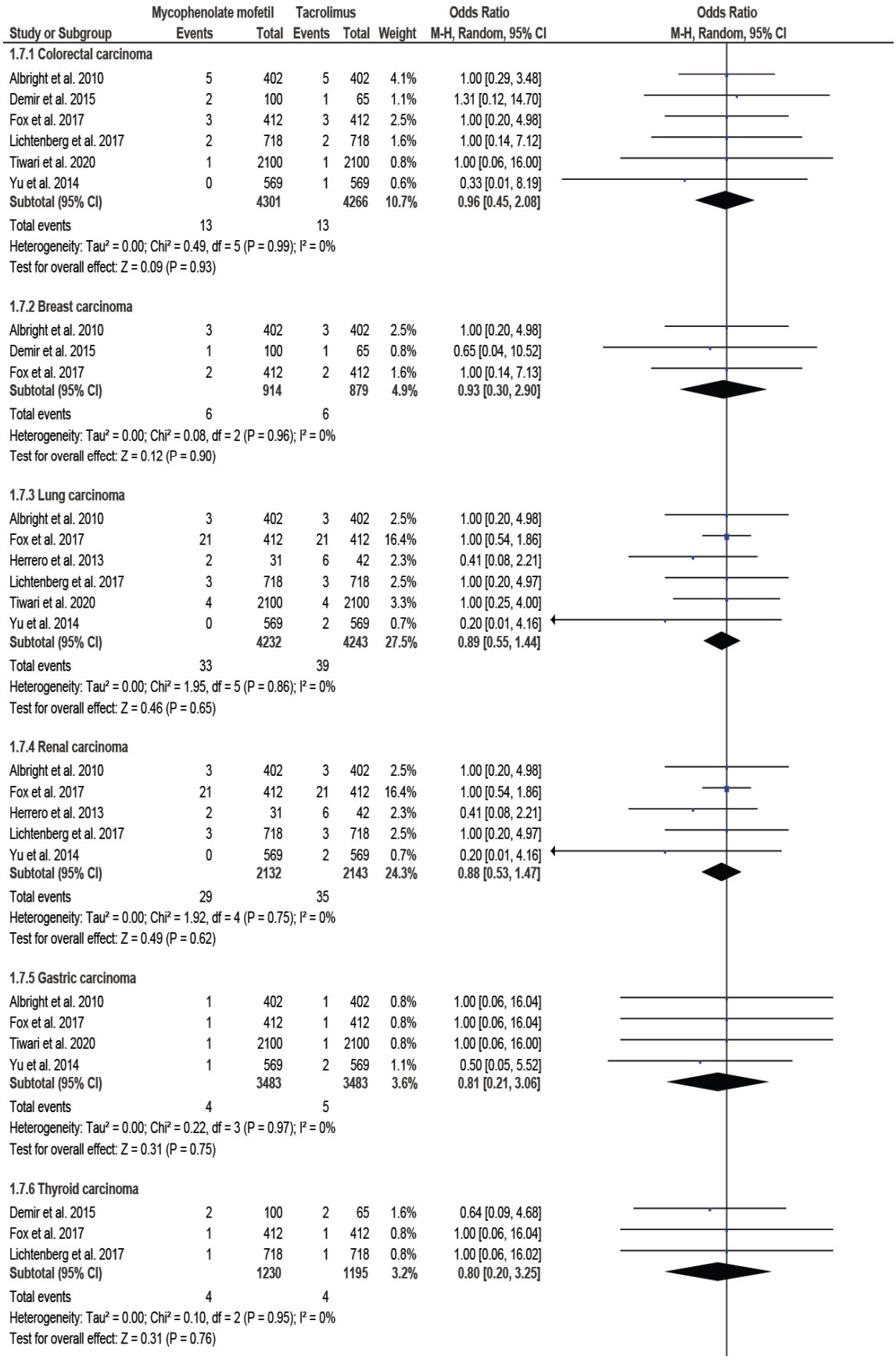


Figure S6. Azathioprine versus Mycophenolate mofetil.


Mycophenolate mofetil
Tacrolimus
Odds Ratio
Odds Ratio
Study or Subgroup Events Total Events Total Weight M-H, Random, 95% CI
M-H, Random, 95% CI
1.7.1 Colorectal carcinoma
Albright et al. 2010 5 402 5 402 4.1% 1.00 [0.29, 3.48]
Demir et al. 2015 2 100 1 65 1.1% 1.31 [0.12, 14.70]
Fox et al. 2017 3 412 3 412 2.5% 1.00 [0.20, 4.98]
Lichtenberg et al. 2017 2 718 2 718 1.6% 1.00 [0.14, 7.12]
Tiwari et al. 2020 1 2100 1 2100 0.8% 1.00 [0.06, 16.00]
Yu et al. 2014 0 569 1 569 0.6% 0.33 [0.01, 8.19]
Subtotal (95% CI) 4301 4266 10.7% 0.96 [0.45, 2.08]
Total events 13 13
Heterogeneity: Tau² = 0.00; Chi² = 0.49, df = 5 (P = 0.99); I² = 0%
Test for overall effect: Z = 0.09 (P = 0.93)
1.7.2 Breast carcinoma
Albright et al. 2010 3 402 3 402 2.5% 1.00 [0.20, 4.98]
Demir et al. 2015 1 100 1 65 0.8% 0.65 [0.04, 10.52]
Fox et al. 2017 2 412 2 412 1.6% 1.00 [0.14, 7.13]
Subtotal (95% CI) 914 879 4.9% 0.93 [0.30, 2.90]
Total events 6 6
Heterogeneity: Tau² = 0.00; Chi² = 0.08, df = 2 (P = 0.96); I² = 0%
Test for overall effect: Z = 0.12 (P = 0.90)
1.7.3 Lung carcinoma
Albright et al. 2010 3 402 3 402 2.5% 1.00 [0.20, 4.98]
Fox et al. 2017 21 412 21 412 16.4% 1.00 [0.54, 1.86]
Herrero et al. 2013 2 31 6 42 2.3% 0.41 [0.08, 2.21]
Lichtenberg et al. 2017 3 718 3 718 2.5% 1.00 [0.20, 4.97]
Tiwari et al. 2020 4 2100 4 2100 3.3% 1.00 [0.25, 4.00]
Yu et al. 2014 0 569 2 569 0.7% 0.20 [0.01, 4.16]
Subtotal (95% CI) 4232 4243 27.5% 0.89 [0.55, 1.44]
Total events 33 39
Heterogeneity: Tau² = 0.00; Chi² = 1.95, df = 5 (P = 0.86); I² = 0%
Test for overall effect: Z = 0.46 (P = 0.65)
1.7.4 Renal carcinoma
Albright et al. 2010 3 402 3 402 2.5% 1.00 [0.20, 4.98]
Fox et al. 2017 21 412 21 412 16.4% 1.00 [0.54, 1.86]
Herrero et al. 2013 2 31 6 42 2.3% 0.41 [0.08, 2.21]
Lichtenberg et al. 2017 3 718 3 718 2.5% 1.00 [0.20, 4.97]
Yu et al. 2014 0 569 2 569 0.7% 0.20 [0.01, 4.16]
Subtotal (95% CI) 2132 2143 24.3% 0.88 [0.53, 1.47]
Total events 29 35
Heterogeneity: Tau² = 0.00; Chi² = 1.92, df = 4 (P = 0.75); I² = 0%
Test for overall effect: Z = 0.49 (P = 0.62)
1.7.5 Gastric carcinoma
Albright et al. 2010 1 402 1 402 0.8% 1.00 [0.06, 16.04]
Fox et al. 2017 1 412 1 412 0.8% 1.00 [0.06, 16.04]
Tiwari et al. 2020 1 2100 1 2100 0.8% 1.00 [0.06, 16.00]
Yu et al. 2014 1 569 2 569 1.1% 0.50 [0.05, 5.52]
Subtotal (95% CI) 3483 3483 3.6% 0.81 [0.21, 3.06]
Total events 4 5
Heterogeneity: Tau² = 0.00; Chi² = 0.22, df = 3 (P = 0.97); I² = 0%
Test for overall effect: Z = 0.31 (P = 0.75)
1.7.6 Thyroid carcinoma
Demir et al. 2015 2 100 2 65 1.6% 0.64 [0.09, 4.68]
Fox et al. 2017 1 412 1 412 0.8% 1.00 [0.06, 16.04]
Lichtenberg et al. 2017 1 718 1 718 0.8% 1.00 [0.06, 16.02]
Subtotal (95% CI) 1230 1195 3.2% 0.80 [0.20, 3.25]
Total events 4 4
Heterogeneity: Tau² = 0.00; Chi² = 0.10, df = 2 (P = 0.95); I² = 0%
Test for overall effect: Z = 0.31 (P = 0.76)

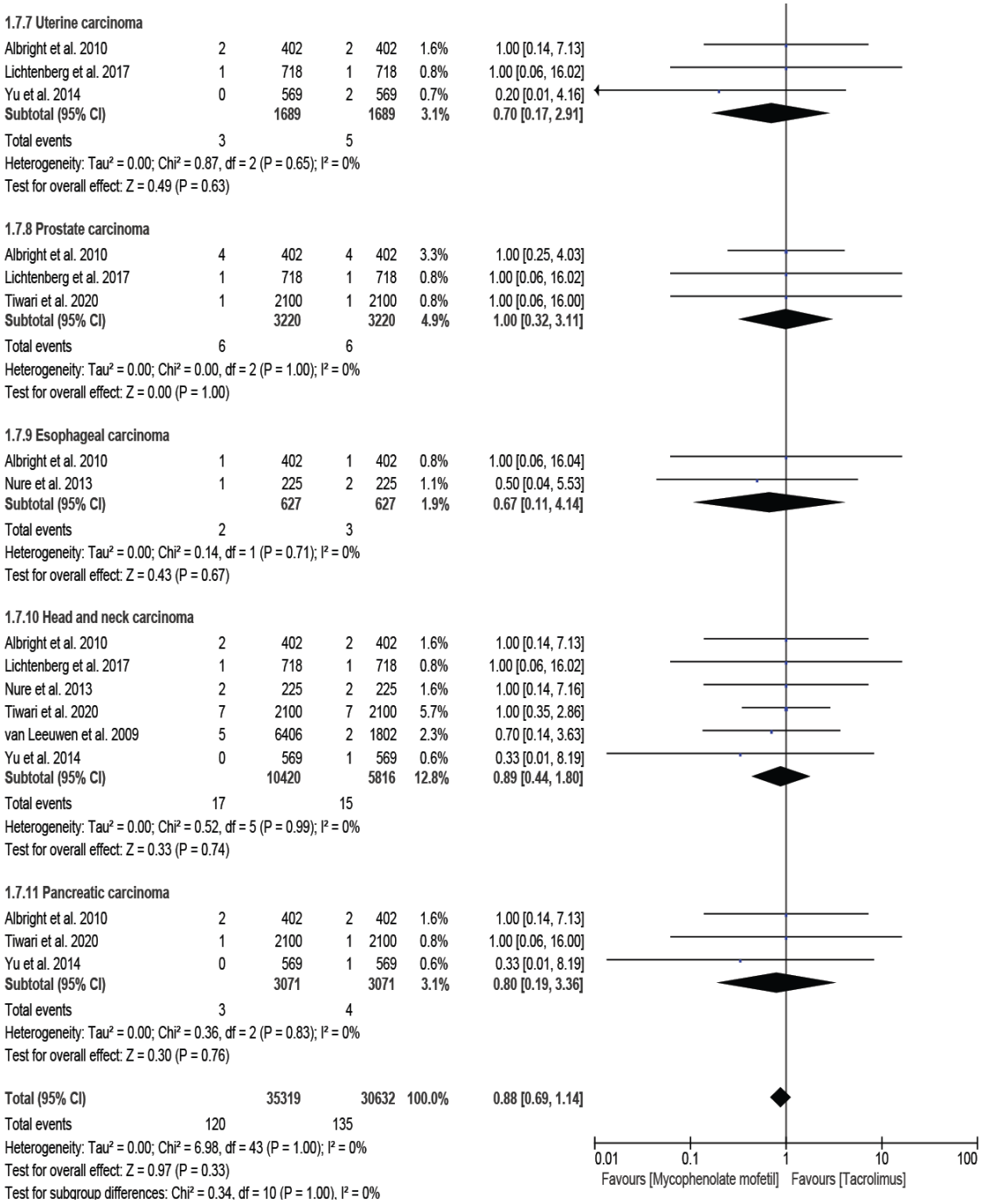


Figure S7. Mycophenolate mofetil versus Tacrolimus.

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